

CPEN 416 / EECE 5710

Lecture 07: superdense coding and quantum teleportation

Thursday 25 September 2025

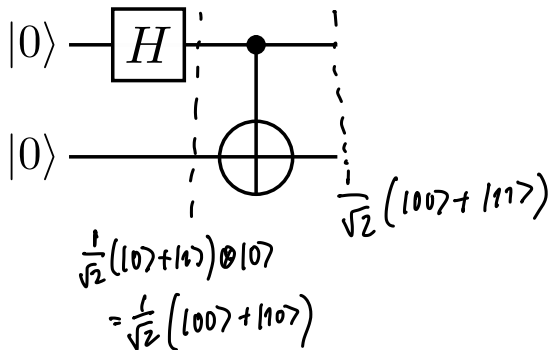
Announcements

post suggested
topics on Piazza

- Tutorial assignment 1 due tomorrow 23:59
- Monday's tutorial: midterm prep
- No class Tuesday, but submit your midterm reference sheets for review

Last time

We introduced the idea of entanglement and entangling gates.



Last time

We saw how applying this circuit to computational basis states creates a basis of entangled states.

$$|\psi_{00}\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$|\psi_{01}\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle)$$

$$|\psi_{10}\rangle = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle)$$

$$|\psi_{11}\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle)$$

Learning outcomes

- measure multi-qubit systems in the Bell basis
- leverage entanglement to implement superdense coding
- prove that arbitrary quantum states cannot be “cloned”
- teleport a state from one qubit to another

Review: basis rotations

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Express in Hadamard basis:
 $|\psi\rangle = \gamma|+\rangle + \delta|-\rangle$

$$\begin{aligned} |\psi\rangle &= \alpha|0\rangle + \beta|1\rangle \\ &= \alpha \cdot \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle) + \beta \cdot \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle) \\ &= \boxed{\frac{1}{\sqrt{2}}(\alpha + \beta)}|+\rangle + \boxed{\frac{1}{\sqrt{2}}(\alpha - \beta)}|-\rangle \end{aligned}$$

meas in $|+\rangle/|-\rangle$ basis:
 $\Pr(+)=\frac{1}{2}|\alpha + \beta|^2$
 $\Pr(-)=\frac{1}{2}|\alpha - \beta|^2$

$$|0\rangle \xrightarrow{C} |\psi\rangle$$

$$\begin{aligned} |+\rangle &= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = H|0\rangle \\ |-\rangle &= \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = H|1\rangle \end{aligned}$$

$$H|+\rangle = |0\rangle$$

$$H|-\rangle = |1\rangle$$

$$|+\rangle + |-\rangle = \frac{2}{\sqrt{2}}|0\rangle = \sqrt{2}|0\rangle$$

$$\begin{aligned} \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle) &= |0\rangle \\ \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle) &= |1\rangle \end{aligned}$$

Review: basis rotations

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$|0\rangle \xrightarrow{C} |\psi\rangle$$

Express in Hadamard basis:
 $|\psi\rangle = \gamma|+\rangle + \delta|-\rangle$

★ Try w/ $|p\rangle / |m\rangle$
yourself

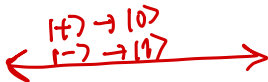
$$\begin{aligned} |\psi\rangle &= \alpha|0\rangle + \beta|1\rangle \\ &= \alpha \cdot \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle) + \beta \cdot \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle) \\ &= \boxed{\frac{1}{\sqrt{2}}(\alpha + \beta)}|+\rangle + \boxed{\frac{1}{\sqrt{2}}(\alpha - \beta)}|-\rangle \end{aligned}$$



$$|\psi\rangle = \frac{1}{\sqrt{2}}(\alpha + \beta)|0\rangle + \frac{1}{\sqrt{2}}(\alpha - \beta)|1\rangle$$

IN HAD. BASIS

$$\begin{aligned} \text{Prob } |+\rangle &= \frac{1}{2}|\alpha + \beta|^2 \\ \text{Prob } |-\rangle &= \frac{1}{2}|\alpha - \beta|^2 \end{aligned}$$



$$\begin{aligned} \text{Prob } |0\rangle &= \frac{1}{2}|\alpha + \beta|^2 \\ \text{Prob } |1\rangle &= \frac{1}{2}|\alpha - \beta|^2 \end{aligned}$$

Bell basis measurement

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \left\{ \begin{array}{c} \text{---} \bullet \text{---} [H] \text{---} |0\rangle \\ | \\ \text{---} \oplus \text{---} |0\rangle \end{array} \right.$$

$$\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \left\{ \begin{array}{c} \text{---} \bullet \text{---} [H] \text{---} |0\rangle \\ | \\ \text{---} \oplus \text{---} |1\rangle \end{array} \right.$$

$$\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \left\{ \begin{array}{c} \text{---} \bullet \text{---} [H] \text{---} |1\rangle \\ | \\ \text{---} \oplus \text{---} |0\rangle \end{array} \right.$$

$$\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \left\{ \begin{array}{c} \text{---} \bullet \text{---} [H] \text{---} |1\rangle \\ | \\ \text{---} \oplus \text{---} |1\rangle \end{array} \right.$$

Two famous quantum algorithms, **superdense coding** and **teleportation** work by performing a measurement in the Bell basis.

Superdense coding

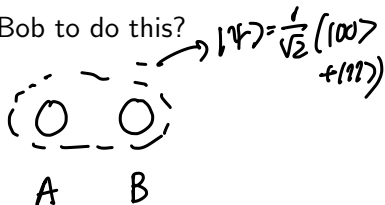
Suppose Alice wants to send Bob two classical bits of information, say '1' and '0'.

Q1: How many classical bits must she send to Bob to do this?

2

Q2: How many *qubits* must she send to Bob to do this?

1



Superdense coding

Alice and Bob start the protocol with this shared entangled state:

$$|\Phi\rangle_{AB} = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

Next, depending on her bits, Alice performs one of the following operations on her qubit:

$$\begin{aligned} 00 &\rightarrow I \\ 01 &\rightarrow X \\ 10 &\rightarrow Z \\ 11 &\rightarrow ZX \end{aligned}$$

Superdense coding

What happened to the entangled state?

$$|\Phi\rangle_{AB} = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

A, B

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

It will transform to:

$$00 \rightarrow I \quad \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$01 \rightarrow X \quad \frac{1}{\sqrt{2}} (|10\rangle + |01\rangle)$$

$$10 \rightarrow Z \quad \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle)$$

$$11 \rightarrow ZX \quad \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle)$$

Superdense coding

Now, Bob can either

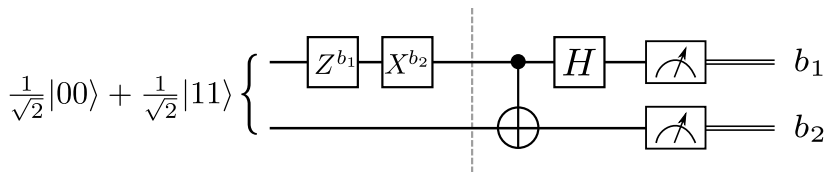
- perform a measurement directly in the Bell basis
- perform a basis transformation from the Bell basis back to the computational basis

to determine his state with certainty, and thus, the bits Alice sent.

$$\begin{aligned}(H \otimes I)\text{CNOT} \cdot \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) &= |00\rangle \\(H \otimes I)\text{CNOT} \cdot \frac{1}{\sqrt{2}} (|10\rangle + |01\rangle) &= |01\rangle \\(H \otimes I)\text{CNOT} \cdot \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle) &= |10\rangle \\(H \otimes I)\text{CNOT} \cdot \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle) &= |11\rangle\end{aligned}$$

Hands-on: superdense coding

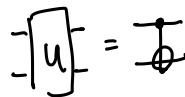
Let's go implement it!



Copying quantum states

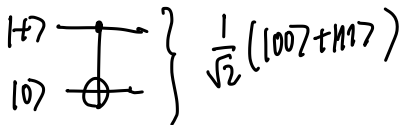
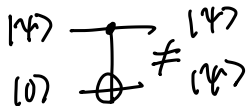
Suppose you found a really cool quantum state, and you want to send a copy to a friend. Under what conditions is this possible?

Exercise: develop a circuit that “copies” $|0\rangle$ and $|1\rangle$.



Copying quantum states

Will your circuit work in general? Is there a state for which it fails?



(Not) copying quantum states

The no-cloning theorem

It is impossible to create a copying circuit that works for arbitrary quantum states.

In other words, there is no circuit that sends

$$|\psi\rangle|s\rangle \mapsto |\psi\rangle|\psi\rangle$$

for any arbitrary $|\psi\rangle$.

Proof of the no-cloning theorem

Suppose we want to clone a state $|\psi\rangle$. We want a unitary operation that sends

where $|s\rangle$ is some arbitrary state.

Suppose we find one. If our cloning machine is to be universal, we must also be able to clone some other state, $|\varphi\rangle$.

Proof of the no-cloning theorem

We purportedly have:

$$U(|\psi\rangle \otimes |s\rangle) = |\psi\rangle \otimes |\psi\rangle$$

$$U(|\varphi\rangle \otimes |s\rangle) = |\varphi\rangle \otimes |\varphi\rangle$$

Take the inner product of the LHS of both equations:

Now take the inner product of the RHS of both equations:

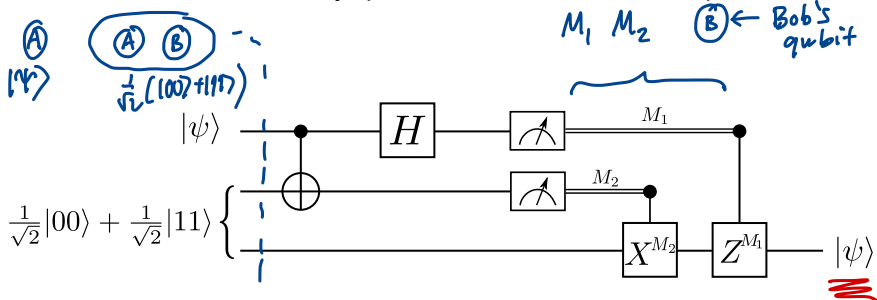
Proof of the no-cloning theorem

For what states does

$$(\langle\psi|\varphi\rangle)^2 = \langle\psi|\varphi\rangle$$

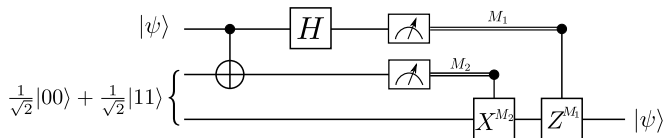
Teleportation

We cannot clone arbitrary qubit states, but we *can* teleport them!



Exercise: determine the state of the system immediately before the measurements.

Teleportation



Quantum teleportation: the details

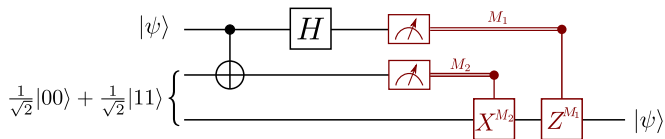
Before measurements, the combined state of the system is

What do you notice about this state?

Quantum teleportation: the details

Alice measures in the computational basis and sends her results to Bob, who can adjust his state as needed.

Let's implement it!



Next time

Content:

- Midterm on 2 October
- After midterm: Oracle-based algorithms (Deutsch and Grover)

Action items:

- 1 Submit tutorial assignment 1
- 2 Study for midterm (practice problems and midterms on PL)
- 3 Submit midterm reference sheet for review by Tuesday 2pm
- 4 Attend Monday's tutorial

Recommended reading (lecture 01-07, and midterm)

- Codebook nodes IQC, SQ, MQ
- Nielsen & Chuang 1.2, 1.3.1-1.3.7, 2.2.3-2.2.5, 2.3, 4.3

Read ahead: Codebook BA, GA; Nielsen & Chuang 1.4, 6.1